

AgriCall: An IoT-Integrated GSM-Based Irrigation Control System for Sustainable Agriculture

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Abstract— This paper proposes *AgriCall*, a cost-effective GSM-based irrigation controller designed to enhance reliability and reduce operational costs for smallholder farmers. The system integrates an Arduino UNO (ATmega328P) with a SIM900A GSM module to enable remote pump switching through missed-call signaling, eliminating call charges. SMS notifications are employed only for critical events such as main power failure or checking status, thereby minimizing communication expenses. Additional features include EEPROM-based user authentication, a 4×4 keypad for stored number management, an LCD for real-time monitoring, and a electricity sensing mechanism with a 5 V backup battery to ensure uninterrupted operation during outages. A functional prototype was developed and tested, demonstrating accurate pump control, timely power-failure alerts, and robust performance under rural operating conditions. The results highlight the potential of the *AgriCall* device to improve irrigation efficiency by enabling timely irrigation control, thereby contributing to water conservation, reduced energy wastage, and enhanced crop productivity in resource-constrained rural environments.

Keywords—Rural automation, GSM-based irrigation, low-cost agricultural technology, calling communication, Remote pump control, IoT in agriculture.

I. INTRODUCTION

The efficiency of irrigation practices constitutes a fundamental element of sustainable agriculture, especially in regions characterized by constrained water resources and limited labor availability. The absence of intelligent control mechanisms in conventional irrigation often prevents timely adaptation to fluctuating power availability and field conditions, necessitating low-cost, communication-driven solutions [1]. Smallholder farmers frequently lack affordable tools for reliable pump management, motivating the development of simple, GSM-based remote irrigation solutions. GSM enables IoT by providing long-range, low-cost cellular connectivity for device-to-device

communication, remote monitoring, and cloud integration. Advancements in GSM-based irrigation systems have introduced richer features such as DTMF-based multi-phase pump control and hybrid IoT cloud solutions for scheduling and analytics, enhancing scalability and resilience [2]. These implementations often depend on SMS, GPRS, or internet-based data exchange, which, despite providing flexibility, also incur recurring costs and increase system complexity.

To address cost concerns, several studies propose missed-call based pump control systems using microcontrollers, valued for their simplicity and low operating expenses. Recent studies published between 2020 and 2025 [3]–[7] emphasize GSM-, SMS-, and IoT-based irrigation systems; however, several limitations persist. Many systems rely heavily on continuous SMS or GPRS communication, resulting in recurring operational costs and higher power consumption. Others lack secure multi-user authentication, local number management, or autonomous fault notification mechanisms.

In contrast, *AgriCall* minimizes communication cost through missed-call control, restricts access using EEPROM-based caller ID authentication, and sends SMS only during critical events such as power failure. This combination directly addresses affordability, reliability, and usability challenges identified in existing systems. Collectively, the literature highlights GSM-enabled microcontrollers as a robust foundation for rural irrigation automation. Compact microcontrollers (ATMEGA328P), notably Arduino Uno, Mega, and UNO, paired with GSM modules, have been applied in irrigation applications involving soil-moisture sensing, pump actuation, and SMS reporting. The literature further identifies the Arduino UNO as a practical option owing to its small form factor, cost efficiency, and GSM compatibility. In this context, the proposed device, *AgriCall* (*Agriculture + Calling system control*), is specifically designed to align with the socio-economic and technological needs of Bangladeshi farmers. By incorporating GSM-based

communication along with IoT-driven data exchange, *AgriCall* facilitates remote control of irrigation pumps, enhances water-use efficiency, and minimizes manual intervention. Existing projects report status via periodic SMS; *AgriCall* design only sends SMS when a critical event (power loss after pump start) occurs, balancing information needs against messaging costs. The system not only eliminating the need for physical presence in the field but also contributes to sustainable agricultural practices by ensuring optimized use of resources. A distinct subset of the literature targets *missed-call* control as a cost-minimizing technique, the device detects incoming call events (caller ID) without requiring the call to be answered, thereby avoiding airtime consumption. *AgriCall* addresses this challenge by combining secure password-protected access, missed-call command handling, and an integrated keypad for local registration and management of authorized numbers, thereby enhancing field usability and security. This study demonstrates how such low-cost, accessible technologies can bridge the gap between traditional irrigation practices and modern precision agriculture.

II. SYSTEM ARCHITECTURE

The *AgriCall* system is developed as a compact IoT-enabled irrigation controller that integrates a GSM communication module with a pump actuation unit. Its architecture is organized into three primary components, ensuring seamless communication, control, and management for field-level irrigation automation:

1. **Control Unit:** This is the central microcontroller-based module that manages the operation of the irrigation pump. It interprets commands received from registered mobile numbers and triggers the pump accordingly.
2. **GSM Communication Module:** The GSM module serves as the communication interface between the farmer's mobile phone and the control unit. It receives SMS or missed-call signals, which are processed by the control unit to execute pump operation.
3. **User Interface & Security Layer:** The system includes a simple keypad interface for registering authorized caller ID and filling password. Only these numbers are allowed to control the pump and ensuring individual operational security.

The device is designed for easy installation, minimal maintenance, and reliable operation in rural agricultural settings. Its modular architecture allows integration with different pump types and varying power supply conditions. A schematic block diagram of the system is represented in Fig. 1.

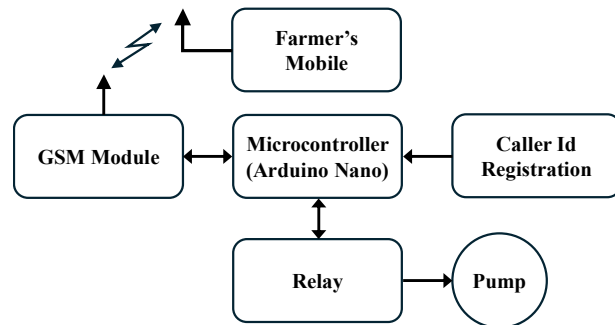


Fig.1. Block diagram of the proposed *AgriCall* model.

III. ACCESSORIES OF THE PROPOSED SYSTEM

A. Microcontroller (Arduino UNO)

The Arduino Uno, based on the ATmega328P 8-bit microcontroller unit (MCU) clock frequency of 16 MHz, serves as the system's core. It provides one SPI, one I2C, and multiple GPIOs, with 32 KB flash, 2 KB SRAM, and 1 KB EEPROM. SPI enables master-slave communication, and the Arduino IDE supports C/C++ programming for embedded control tasks [8].

B. GSM Module

The GSM SIM900A Mini is a compact cellular module operates at 3.2–4.8 V and supporting dual-band GSM/GPRS (900, 1800 MHz) for voice, SMS, and data transmission. It interfaces with microcontrollers via UART, supports AT command control, and enables remote communication and suitable for embedded IoT applications [9].

C. Liquid Crystal Display (LCD)

The 16×2 LCD is a display module capable of showing 16 characters per line on 2 lines. It typically uses the HD44780 controller, supports 4-bit or 8-bit parallel interfacing with microcontrollers, and can be operated in character or command mode. Backlight and contrast are adjustable for embedded applications [10].

D. Keypad

The 4×4 matrix keypad has 16 keys arranged in 4 rows and 4 columns. It interfaces with microcontrollers via row-column scanning, providing digital input for password entry, control commands, or menu navigation in embedded systems.

E. Relay

A relay module is an electromechanical switch controlled by a microcontroller to operate high-voltage AC/DC loads. It provides isolation via an optocoupler, supports NO/NC contacts, and can include electricity sensing for monitoring load status in embedded systems.

The Arduino UNO paired with the SIM900A GSM module was selected over a GPRS-based solution due to its lower power consumption, reduced system complexity, and elimination of recurring data costs. While GPRS enables continuous data transmission, it depends on reliable internet availability and incurs higher energy usage and subscription expenses. In contrast, the SIM900A supports call- and SMS-based communication, which is more dependable in rural regions with limited data connectivity. The Arduino UNO is utilized primarily for prototyping due to its ease of programming, extensive community support, and rapid development capability. For large-scale deployment, the system can be implemented on a custom PCB using the standalone ATmega328P microcontroller, thereby reducing cost, size, and power consumption while maintaining identical functionality.

IV. PROJECT DESIGN AND WORKING PRINCIPLE

A schematic of the proposed *AgriCall* system is shown in Fig. 2, illustrating a GSM-based irrigation solution for remote operation of AC pumps. The system employs the ATMEGA328P microcontroller (Arduino UNO) as the central controller for command execution, relay actuation, and peripheral coordination. The GSM SIM900A module, interfaced via UART (TX–D7, RX–D8), detects missed calls

and signals the controller without establishing a voice connection, eliminating airtime or data usage. User authentication is managed through a 4×4 matrix keypad (rows A0–A3, columns 4, 5, 6, 10), enabling secure registration of up to three caller IDs with password protection. Functional keys provide local control: ‘#’ stores a number, ‘D’ deletes, ‘C’ retrieves, ‘B’ manually activates the pump, and ‘A’ resets the system. A 16×2 LCD, interfaced via I2C (SDA–A4, SCL–A5), displays real-time alerts and operational status. The relay, driven from pin D9 through a transistor, switches the AC pump using a low-voltage signal while isolating the controller from surges. Mains availability is monitored via pin D3; if absent, the Arduino commands the GSM module to issue an SMS alert, ensuring reliable pump monitoring and timely farmer notifications.

During operation, a registered user initiates a missed call to the GSM module, which subsequently signals the Arduino UNO. The caller’s id, pre-stored in EEPROM during the registration phase, is authenticated via caller ID verification. Upon successful validation, the Arduino actuates the relay to control the pump (start/stop). Concurrently, the relay sensing circuit continuously monitors the availability of electrical power. On the other side pump deactivation is achieved through multiple controlled methods. A registered user can turn the pump OFF by issuing another authenticated missed call, which toggles the relay state. Manual pump control is also available through the local keypad interface by pressing ‘*’ for emergency situations. Furthermore in the event of a power outage following pump activation, the Arduino immediately instructs the GSM module to transmit an SMS alert to the farmer, thereby ensuring real-time awareness of pump operation and power supply conditions without the need for manual supervision. This multi-layered shutdown

mechanism ensures operational safety, flexibility, and protection of irrigation equipment.

The proposed system integrates microcontroller-based control, GSM communication, keypad input, I2C-based LCD monitoring, relay actuation, and power sensing to establish a cost-effective, reliable, and energy-efficient irrigation automation solution. By providing secure access, automated pump control, real-time alerts, and power monitoring, the *AgriCall* system minimizes human intervention, improves operational efficiency, optimizes water utilization, and contributes to sustainable agricultural practices in rural settings. The complete implementation of the proposed system is presented in Fig. 3, demonstrating real-time operation. Fig. 4 depicts the interfacing of (a) registered numbers, (b) incoming calls, and (c) pump activation, whereas Fig. 5 illustrates (a) mobile call snapshots and (b) mobile SMS snapshots. The control logic is programmed on the Arduino platform using the C language, where the code defines a sequence of instructions to be executed by the microcontroller. During supply interruptions, the Arduino communicates with the GSM module to issue an SMS alert. Additionally, farmers can remotely query pump status using concise text commands (e.g., “status?”), to which the system responds with the current load condition, thereby eliminating the necessity of physical inspection. To enhance reliability, a DC backup battery ensures uninterrupted operation during mains failure, remaining in charging mode under normal conditions. The complete functional sequence, including decision-making and signal processing stages, is summarized in the flowchart of Fig. 6, demonstrating the effectiveness of integrating embedded control with low-cost communication technologies for smart agricultural applications.

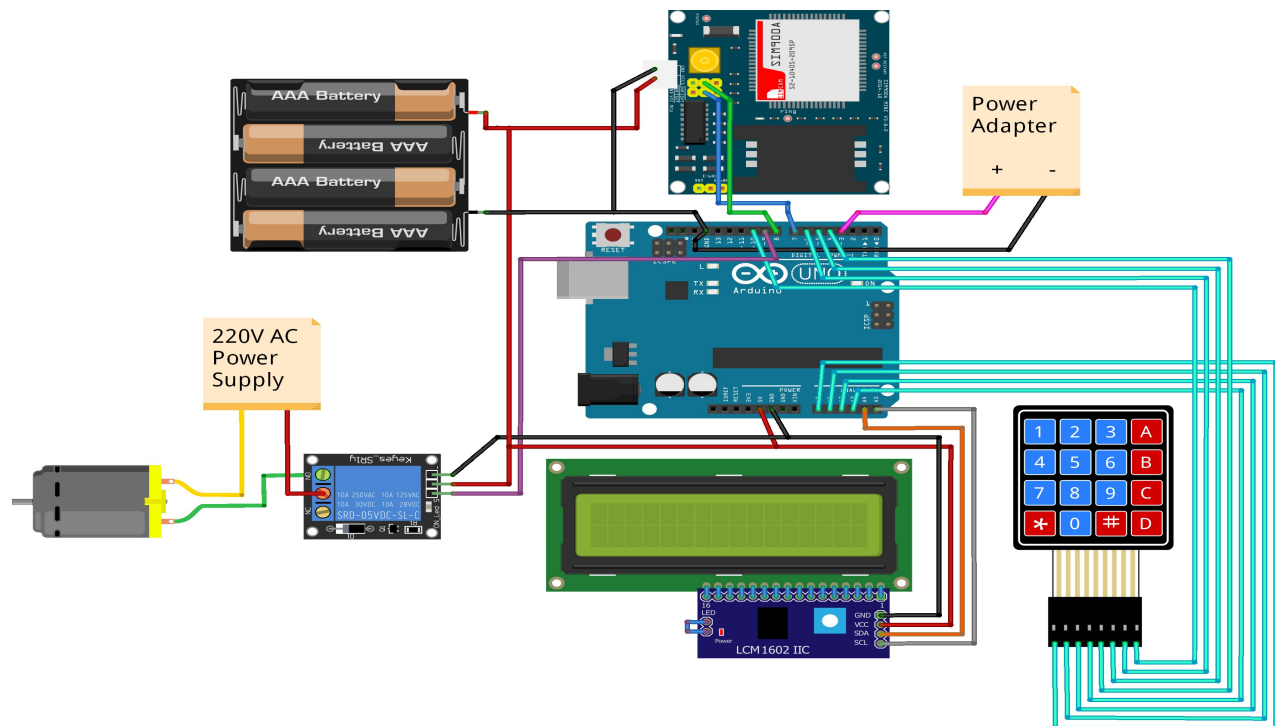


Fig. 2: Circuit configuration of the proposed *AgriCall* system.

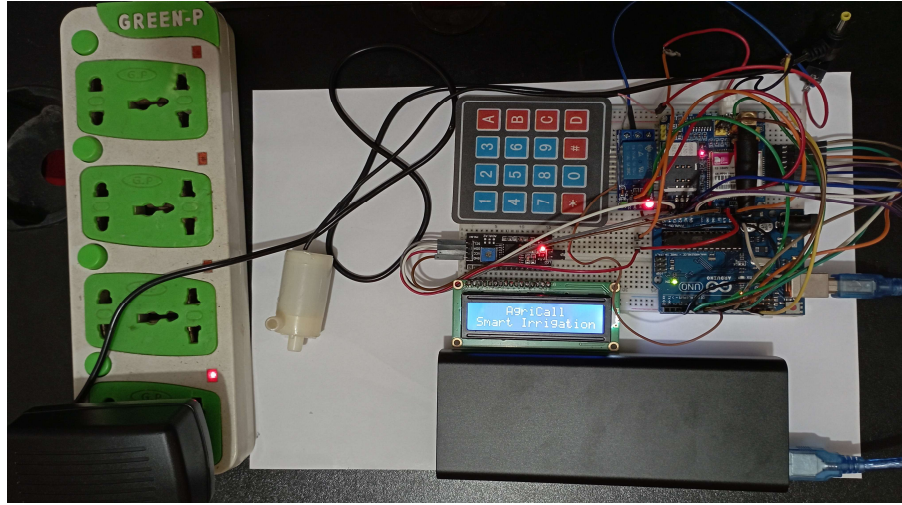


Fig. 3: Real-time implementation of the proposed *AgriCall* system.

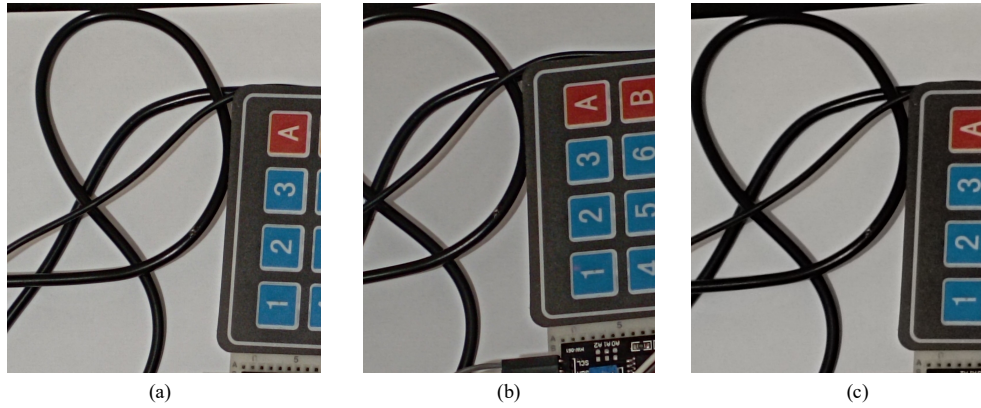


Fig. 4: System interfacing feature with (a) registered numbers, (b) incoming calls, and (c) load shedding.

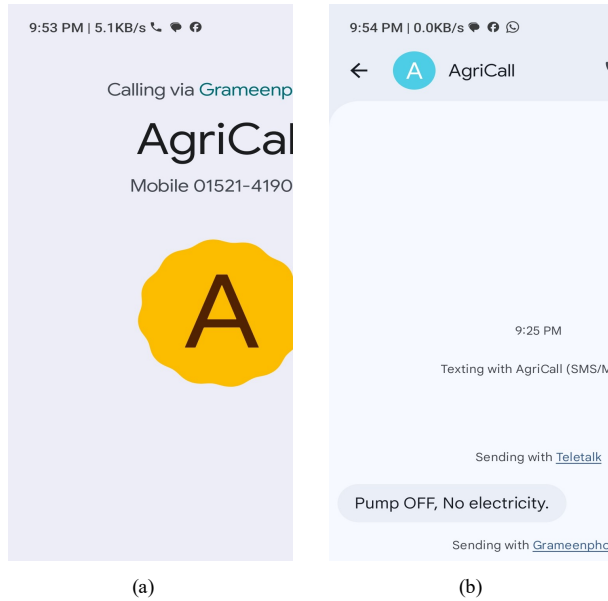


Fig. 5: Mobile interfacing (a) incoming calls and (b) Responding message from system.

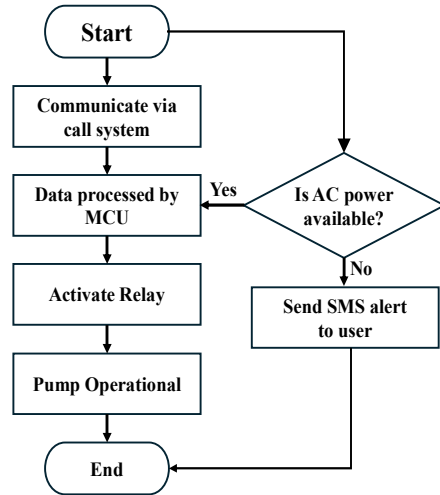


Fig. 6: Operational flow diagram of the proposed *AgriCall* system.

V. CONCLUSION

Experimental evaluation demonstrates that *AgriCall* offers high operational reliability, cost-effectiveness, and consistent performance under rural power and network conditions for smallholder farmers. Unlike conventional systems that rely heavily on frequent SMS communication, the missed-call control strategy eliminates recurring

communication costs, while EEPROM-based authentication ensures secure access. Field evaluation confirmed the system's ability to deliver prompt fault notifications and maintain essential functionality during power outages, highlighting its suitability for rural deployment. Beyond cost savings, the approach enhances farmer convenience and lays the groundwork for more resilient agricultural practices. Future work will focus on enhancing the AgriCall system by integrating soil-moisture, adoption of renewable power options to improve system autonomy, and environmental sensors to enable demand-driven, automated irrigation decisions. For large-scale deployment, the current prototype implemented using an Arduino UNO will be transitioned to a custom PCB design based on the standalone ATmega328P microcontroller. This approach will significantly reduce system size, power consumption, and production cost, while improving reliability and scalability for mass manufacturing. These findings position *AgriCall* as a viable, affordable, and scalable irrigation automation solution for smart and sustainable agriculture, particularly for smallholder farming communities.

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